

TEVIX — Research Sources & Outcomes

Companion bibliography to **TEVIX_Feasibility_Report.pdf**. Every source from two fact-checked research passes, with a one-line brief and the specific outcome for TEVIX. Quality tags: **PRIMARY** peer-reviewed / government / legal · **SECONDARY** industry/news/aggregator · **LOW** unreliable aggregator. One row is marked REFUTED — listed so the team does NOT cite it.

A · Sensing science & biosensors

Source & link	What it is	Outcome for TEVIX
Frontiers in Sustainable Food Systems 2026 PRIMARY doi.org/10.3389/fsufs.2026.1814448	Review of biosensors for food spoilage & pathogen detection	Electrochemical/optical/piezoelectric biosensors detect pathogens at single-digit CFU/mL; chemiresistive/electrochemical are the most embeddable class for low-power wireless packaging.
PMC12346877 PRIMARY pmc.ncbi.nlm.nih.gov/articles/PMC12346877	Review of pathogen biosensor detection times & commercialization	Detection in 20–80 min (E. coli 30 min, Salmonella 25 min, Listeria 60 min). Commercialization still limited; existing products are benchtop, not embedded.
PMC12385247 PRIMARY pmc.ncbi.nlm.nih.gov/articles/PMC12385247	Review of colorimetric food-freshness indicators (CFFIs)	CFFIs detect TVB-N, VOCs, H ₂ S, biogenic amines, CO ₂ , pH. Key cost fact: smart materials = 50–100% of packaging cost vs <10% viability target. Mostly lab-stage (only RipeSense, SensorQ, FreshTag commercial).
ScienceDirect S2666154326001201 PRIMARY sciencedirect.com/science/article/pii/S2666154326001201	Review: AI + gas-sensor arrays + IoT in smart packaging	Validates the AI pillar: IoT+ML enables dynamic shelf-life prediction; TinyML on Arduino hit 91.9% accuracy. Limits = sensor stability, cost, non-biodegradable substrates.
Wiley Food Bioengineering 10.1002/fbe2.70035 SECONDARY onlinelibrary.wiley.com/doi/10.1002/fbe2.70035	Review of food biosensor engineering	Corroborates commercialization barriers: cost, stability, regulatory validation, standardization.

B · MEMS resistive sensing & analog readout (evaluator focus)

Source & link	What it is	Outcome for TEVIX
PMC9866592 PRIMARY pmc.ncbi.nlm.nih.gov/articles/PMC9866592	CMOS readout IC for resistive MEMS gas sensor	100 Ω–1 MΩ (4 decades), 0.5–1% error, Wheatstone bridge + chopper-stabilized VGA + 12-bit SAR ADC, 0.98 mW. Caveat: post-layout simulation, not silicon.
MDPI Sensors 18(3):761 PRIMARY mdpi.com/1424-8220/18/3/761	Fabricated reconfigurable readout IC (0.18 μm)	Selectable RDC/12-bit SAR/16-bit ΔΣ; dual-mode front-end 250 Ω–10 MΩ (~4.6 decades), resolution-vs-power tradeoff.
ScienceDirect S0924424725001529 PRIMARY sciencedirect.com/science/article/abs/pii/S0924424725001529	2025 readout IC, range extension	Doubles single-slope ADC input range via extra MSB phase — wide-range read technique.

MDPI Micromachines 16060658 PRIMARY doi.org/10.3390/mi16060658	2025 CMOS-SOI-MEMS GMOS voltage-mode readout	Three-terminal active/blind differential pair, 5 μ A/branch, 27 dB gain, correction DAC, inst-amp gains 6–42 dB. Concrete low-power signal-conditioning design.
Springer chapter 978-3-642-21538-4_36 PRIMARY link.springer.com/chapter/10.1007/978-3-642-21538-4_36	Chopper-stabilized amplifier for high-resistance sensors	Chopper stabilization is THE technique to suppress DC offset + 1/f flicker noise when reading high-resistance MEMS sensors. (Foundational ~2011, still standard.)
ResearchGate 221375068 REFUTED researchgate.net/publication/221375068	Claimed 160 dB dynamic-range resistance-to-time converter	DO NOT CITE. The "160 dB / 0.1% over 5 decades" claim failed verification (1–2 vote).

C · Power, NFC & printed electronics

Source & link	What it is	Outcome for TEVIX
MDPI Sensors 19(7):1741 (Lazaro) PRIMARY mdpi.com/1424-8220/19/7/1741	Battery-less NFC colour-sensor tag, full design + BOM	BOM < \$5 at volume (colour sensor \$1.5 + NFC IC \$0.3 + MCU \$1.0). Draws 3 mA < 5 mA harvestable → phone-powered, no battery. Read range ~2 cm.
Nature Sci. Reports 2019 s41598-019-53775-1 PRIMARY ncbi.nlm.nih.gov/pmc/articles/PMC6879496	Battery-free NFC food-spoilage monitoring tag	Tag consumes <1 mW (vs >10 mW ordinary); smartphone supplies several mW — fully battery-less. Demo used a spoilage-proxy modality.
MDPI Sensors 2018 survey PRIMARY ncbi.nlm.nih.gov/pmc/articles/PMC6263776	NFC energy-harvesting survey	Passive NFC at 13.56 MHz harvests ~5 mA at 2–3 V (advanced ICs up to 10 mA). Confirms phone-powered feasibility, distance-limited.
Future Foods 2022 PRIMARY pmc.ncbi.nlm.nih.gov/articles/PMC9576266	Peer-reviewed RFID cost table for food packaging	Passive UHF RFID \$0.09–0.18/unit at 100k; price drops 17–33% from 10k→100k units.
ACS Sensors 2022, Barandun (Imperial) PRIMARY pubs.acs.org/doi/10.1021/acssens.2c01086	Printed gas sensor for raw poultry packaging	The viability threshold: printed sensor must add ≤ ~US ¢1/unit. The hard cost target TEVIX must engineer toward.
AllAboutCircuits / Tageos FlexIC SECONDARY allaboutcircuits.com/news/tageos-unveils-flexic-based-nfc-inlays-using-pragmatic-semis-paper-chip	News: paper/flexible NFC inlays	Signals printed/flexible NFC trajectory toward lower cost — directional support for cost-down path.

D · India food loss, exports & market

Source & link	What it is	Outcome for TEVIX
ICRIER PDF (ICAR-CIPHET 2015, Jha et al.) PRIMARY icrier.org/pdf/Reducing-post-harvest-losses-in-India.pdf	Govt post-harvest loss study, 46 commodities	Total loss ₹92,651 cr ≈ USD 17.1 bn (2012–13); ~77–87% at farm level. The headline problem-size number.

PIB PRID 1885038 (MoFPI / NABCONS 2022) PRIMARY pib.gov.in/PressReleaseFramePage.aspx?PRID=1885038	Latest pan-India per-commodity loss %	Fruits 6–15%, marine fish 8.76%, vegetables 4.9–11.6%, meat 2.34%, milk 0.87%. Identifies perishables as TEVIX's target loss categories.
PIB PRID 2006731 (Min. of Commerce 2024) PRIMARY pib.gov.in/PressReleasePage.aspx?PRID=2006731	India agri-export value	USD 53.1 bn agri exports (FY2022–23); APEDA USD 26.7 bn. Export-quality is a real driver.
IBEF / PHDCCI SECONDARY ibef.org/news/indian-food-processing-industry-to-reach...700-billion-by-2030	Food-processing industry projection	USD 307 bn (2023) → USD 700 bn by 2030 (~12.5% CAGR). Directional TAM only.
Precedence Research LOW precedenceresearch.com/smart-packaging-market	Aggregator market report	Global smart packaging \$32.2 bn (2025) → \$55.6 bn (2034), 6.24% CAGR. Unverified — directional.
GMIInsights LOW gminsights.com/industry-analysis/active-and-intelligent-packaging-market	Aggregator market report	Active & intelligent packaging \$14.9 bn (2025) → \$41.4 bn (2035), 11% CAGR. Unverified.
MarketsandMarkets SECONDARY marketsandmarkets.com/PressReleases/biosensors.asp	Aggregator biosensor report	Global biosensors \$34.5 bn (2025) → \$54.4 bn (2030), 9.5% CAGR. Directional.

E • Regulatory (primary legal texts)

Source & link	What it is	Outcome for TEVIX
EUR-Lex Reg (EC) 450/2009 PRIMARY eur-lex.europa.eu/eli/reg/2009/450/oj/eng	EU active & intelligent packaging regulation	Defines "intelligent" materials (monitor food condition — TEVIX fits); must not release into food; only Community-list substances allowed.
EUR-Lex Reg (EU) 10/2011 PRIMARY eur-lex.europa.eu/eli/reg/2011/10/oj/eng	EU plastics food-contact regulation	Overall Migration Limit 60 mg/kg (10 mg/dm²), specific migration limits, Union List, Declaration of Compliance.
FSSAI Packaging Compendium V (Apr 2025) PRIMARY fssai.gov.in/upload/uploadfiles/files/Compendium_Packaging_V_02-04-2025.pdf	India's in-force packaging regulation	OML 60 mg/kg per IS 9845; SMLs (Ba 1.0, Co 0.05, Zn 25.0, DEHP 1.5). Critically: ZERO provisions for smart/active/intelligent/sensors — regulatory vacuum. rPET allowed since 28 Mar 2025.
EC Food-Contact Materials legislation PRIMARY food.ec.europa.eu/food-safety/chemical-safety/food-contact-materials/legislation_en	EU framework Reg 1935/2004 portal	Food-contact materials must not endanger health or alter food — baseline TEVIX sensor must meet.
FDA Food-Contact Substances PRIMARY fda.gov/food/packaging-food-contact-substances-fcs	US FDA FCS/FCN regime	US pathway exists via FCN; intelligent-packaging-specific requirements not yet detailed (gap).

Keller & Heckman / packaginglaw.com SECONDARY packaginglaw.com/news/india-amendments-regulations-allow-use-recycled-pet-fcms	Legal analysis of India rPET amendment + draft 2026 rules	Confirms draft 2026 FSSAI amendment adding "active and intelligent materials" is only in comment period — not in force.
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F • Competitors & market commentary

Source & link	What it is	Outcome for TEVIX
The Spoon SECONDARY thespoon.tech/evigence-raises-18m-for-its-food-freshness-sensors...	News on Evigence funding	Evigence raised USD 18M for sticker-sized freshness sensors — a funded direct-adjacent competitor.
StartUs Insights SECONDARY startus-insights.com/innovators-guide/discover-5-top-sensor-embedded-packaging-solutions	5 sensor-embedded packaging startups	Surfaces Senoptica (O ₂ colorimetric) & Blakbear (gas label + Freshness API = closest TEVIX competitor); others are logistics/anti-counterfeit, not freshness.
DigiComply SECONDARY digicomply.com/blog/embedded-spoilage-indicators-smart-packaging-moves-inside-the-product	Overview of embedded spoilage indicators	Maps the "sense-but-don't-predict" gap TEVIX exploits.
Avery Dennison RFID SECONDARY rfid.averydennison.com/en/home/news-insights/insights/food-bringing-technology-into-packaging.html	Industry view on RFID in food packaging	RFID-in-packaging is commercial but tracking-focused, not freshness-predictive.

Generated from two adversarially fact-checked research passes (218 total agents, ~3.6M tokens). 48 of 50 verified claims confirmed; 2 refuted and excluded. Aggregator market figures retained only as directional context with explicit LOW-quality tags.